

The double-cone Sobolev interpolation question was answered by Auscher–Badr

Literature-status note for arXiv:0705.2216

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Status. Literature already answered. This note identifies an explicit later answer; it does not claim a new theorem.

Source question. Let

$$X = \{(x_1, \dots, x_n) \in \mathbb{R}^n : x_1^2 + \dots + x_{n-1}^2 \leq x_n^2\}, \quad \Omega = \text{int } X,$$

the union of two closed cones meeting only at the vertex. For finite p , Badr defines $H_p^1(X)$ as the closure of $\text{Lip}_0(X)$ for

$$\|f\|_{H_p^1(X)} = \|f\|_{L^p(\Omega)} + \|\nabla f\|_{L^p(\Omega)}.$$

The Appendix of *Real interpolation of Sobolev spaces*, arXiv:0705.2216, asks on printed page 23 whether the H_p^1 interpolate for all $1 \leq p \leq \infty$.

Exact later answer. Pascal Auscher and Nadine Badr study the same double cone and the same spaces in *Sobolev spaces on multiple cones*, arXiv:0812.1146. Their Theorem 3.2 (printed page 5) states that, whenever

$$1 \leq p_0 < p < p_1 \leq \infty, \quad \frac{1}{p} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1},$$

one has, with equivalent norms,

$$(H_{p_0}^1(X), H_{p_1}^1(X))_{\theta, p} = \begin{cases} H_p^1(X), & p \neq n, \\ \widehat{H}_n^1(X), & p = n. \end{cases}$$

Here, if f_r is the spherical radial mean of f and $f_a = f - f_r$ is its anti-radial part, then

$$\widehat{H}_n^1(X) = \left\{ f \in H_n^1(X) : \frac{f_a}{|x|} \in L^n(X) \right\}, \quad \|f\|_{\widehat{H}_n^1} = \|f\|_{H_n^1} + \left\| \frac{f_a}{|x|} \right\|_{L^n}.$$

The paper proves that $\widehat{H}_n^1(X)$ is a strict dense subspace of $H_n^1(X)$. For example, in dimension two, a compactly supported function equal near the vertex to $\text{sign}(x_2)|\log|x||^{-\beta}$ lies in $H_2^1(X)$ for every $\beta > 0$, but lies in $\widehat{H}_2^1(X)$ only when $\beta > 1/2$.

Conclusion. The answer to the source question is *no*: the family $(H_p^1(X))_{1 \leq p \leq \infty}$ fails to be a real interpolation scale at the single critical exponent $p = n$. The later paper also gives the exact replacement space at that exponent. Its introduction explicitly identifies the previously unsettled multiple-cone question as its motivation.

References. N. Badr, *Real interpolation of Sobolev spaces*, Math. Scand. 105 (2009), 235–264; arXiv:0705.2216. P. Auscher and N. Badr, *Sobolev spaces on multiple cones*, Ann. Fac. Sci. Toulouse Math. 19 (2010), 707–733; arXiv:0812.1146; DOI: 10.5802/afst.1264.